

NAME: THAMIZHARASAN M

ROLL NO: 240701560

EXPERIMENT – 06

Deluge Scripting Basics - Learning Simulator

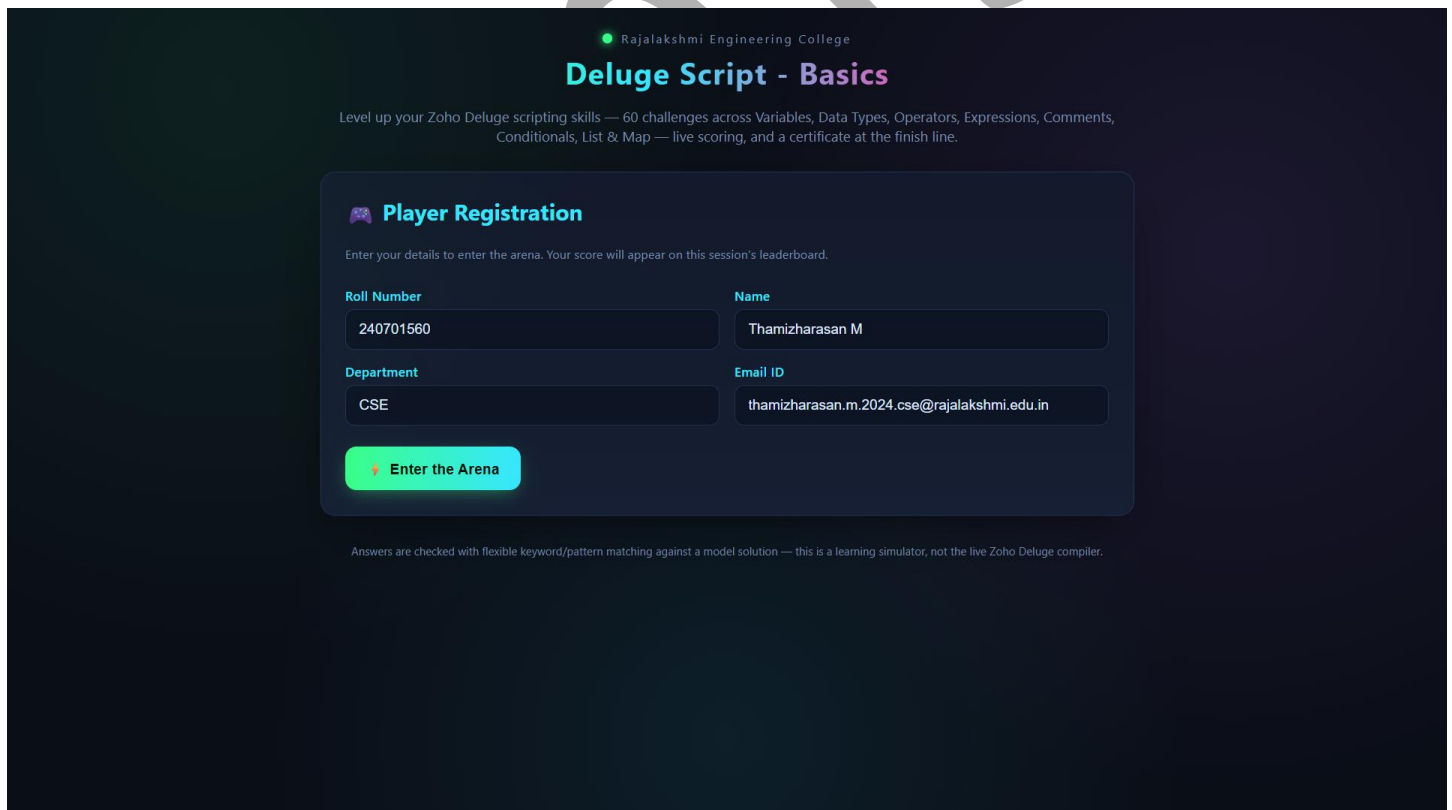
Aim:

To master the fundamentals of the Zoho Deluge scripting language by successfully completing a series of interactive programming challenges covering variables, data types, operators, expressions, conditional statements, lists, and maps using the learning simulator platform.

Procedure:

1. 1. Simulator Registration:

Enter the Deluge Script learning arena by registering with the designated Roll Number (240701560), Name (Thamizharasan M), Department (CSE), and Email ID.



The screenshot shows the 'Deluge Script - Basics' registration interface. At the top, it says 'Rajalakshmi Engineering College' and 'Deluge Script - Basics'. Below this, a description states: 'Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.' The main section is titled 'Player Registration' with a game controller icon. It instructs the user to 'Enter your details to enter the arena. Your score will appear on this session's leaderboard.' There are four input fields: 'Roll Number' (240701560), 'Name' (Thamizharasan M), 'Department' (CSE), and 'Email ID' (thamizharasan.m.2024.cse@rajalakshmi.edu.in). A green 'Enter the Arena' button is at the bottom. A small disclaimer at the very bottom reads: 'Answers are checked with flexible keyword/pattern matching against a model solution — this is a learning simulator, not the live Zoho Deluge compiler.'

2. 2. Scripting Fundamentals - Comments:

Implement both single-line (`//`) and multi-line (`/* */`) comments in Deluge syntax to annotate code effectively.

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Deluge Script - Basics

Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.

Thamizharasan M

Score: 1.6 /100

Streak: 1

Q 1 / 60

Comments

MEDIUM

Write a multi-line comment (using `/*` and `*/`) that says: `This function calculates the result based on marks` .

Your Deluge Script

copy / paste / right-click disabled

```
/*
This function calculates the result based on marks
*/
```

Correct! +1.6 points. Great job on Comments.

Model answer: `/* This function calculates the result based on marks */`

Next Question →

Answers are checked with flexible keyword/pattern matching against a model solution — this is a learning simulator, not the live Zoho Deluge compiler.

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Thamizharasan M

Score: 9.6 /100

Streak: 5

Q 5 / 60

Comments

EASY

Write a single-line comment in Deluge explaining: `this calculates total marks` .

Your Deluge Script

copy / paste / right-click disabled

```
// this calculates total marks
```

Correct! +1.1 points. Great job on Comments.

Model answer: `// this calculates total marks`

Next Question →

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3. 3. Variables and Data Types:

Declare variables holding String data types (e.g., `department = "AI & DS";`) and perform variable reassignment (e.g., updating a `score` variable to 20).

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Thamizharasan M

Score: 20.2 / 100

Streak: 12

Q 12 / 60

Data Types

EASY

Declare a variable `department` holding a Text (String) data type value `"AI & DS"`.

Your Deluge Script copy / paste / right-click disabled

```
department = "AI & DS";
```

Correct! +1.1 points. Great job on Data Types.

Model answer: `department = "AI & DS";`

Next Question →

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Thamizharasan M

Score: 10.6 / 100

Streak: 6

Q 6 / 60

Variables

EASY

A variable `score` already holds 10. Write a line that reassigns its value to `20`.

Your Deluge Script copy / paste / right-click disabled

```
score = 20;
```

Correct! +1.1 points. Great job on Variables.

Model answer: `score = 20;`

Next Question →

Answers are checked with flexible keyword/pattern matching against a model solution — this is a learning simulator, not the live Zoho Deluge compiler.

4. 4. Operators and Expressions:

Construct mathematical expressions using operators (e.g., modulus for checking even numbers) and logical expressions using AND (`&&`) operators to evaluate multiple conditions.

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Thamizharasan M

Score: 18.1 / 100

Streak: 10

Q 10 / 60

Expressions

MEDIUM

Write an expression that checks whether `num` is even, storing the boolean result in `isEven`, using the modulus operator.

Your Deluge Script copy / paste / right-click disabled

```
isEven = num % 2 == 0;
```

Correct! +1.6 points. Great job on Expressions.

Model answer: `isEven = (num % 2 == 0);`

Next Question →

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Thamizharasan M

Score: 16.5 /100

Streak: 9

Q 9 / 60

Operators

MEDIUM

Write a logical expression using AND (`&&`) that checks `age` is greater than 18 and `hasID` is true, storing result in `canVote` .

Your Deluge Script

`canVote = age > 18 && hasID;`

Correct! +1.6 points. Great job on Operators.

Model answer: `canVote = (age > 18 && hasID == true);`

Next Question →

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Thamizharasan M

Score: 3.2 /100

Streak: 2

Q 2 / 60

Expressions

MEDIUM

Write an expression using parentheses to correctly compute `(a + b) * c` , storing the result in `output` .

Your Deluge Script

`output = (a + b) * c;`

Correct! +1.6 points. Great job on Expressions.

Model answer: `output = (a + b) * c;`

Next Question →

Answers are checked with flexible keyword/pattern matching against a model solution — this is a learning simulator, not the live Zoho Deluge compiler.

5. 5. Conditional Logic:

Write conditional flow control statements, including simple `if` blocks and complex `if-else-if-else` chains to categorize data based on specific age criteria.

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Thamizharasan M

Score: 5.9 /100

Streak: 3

Q 3 / 60

Conditional Statements

HARD

Write an if-else-if-else chain that assigns `category` based on `age` : under 13 -> "Child", under 20 -> "Teen", else -> "Adult".

Your Deluge Script copy / paste / right-click disabled

```
category = "Teen";
}
else
{
    category = "Adult";
}
```

Correct! +2.7 points. Great job on Conditional Statements.

Model answer: `if(age < 13) { category = "Child"; } else if(age < 20) { category = "Teen"; } else { category = "Adult"; }`

Next Question →

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Thamizharasan M

Score: 19.1 /100

Streak: 11

Q 11 / 60

Conditional Statements

EASY

Write an if statement that sets `message` to "Adult" when `age` is greater than or equal to 18.

Your Deluge Script copy / paste / right-click disabled

```
if (age >= 18)
{
    message = "Adult";
}
```

Correct! +1.1 points. Great job on Conditional Statements.

Model answer: `if(age >= 18) { message = "Adult"; }`

Next Question →

Answers are checked with flexible keyword/pattern matching against a model solution — this is a learning simulator, not the live Zoho Deluge compiler.

6. 6. Data Collections - Lists & Maps:

Initialize and manipulate collection data types. Create Lists and use the `.add()` function to append values. Interact with Maps by using `.containsKey()` to check for the existence of a key and `.get()` to retrieve specific values.

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Thamizharasan M | Score: 13.3 / 100 | Streak: 7 | Q 7 / 60

Map

Check whether the map `student` contains the key `"email"`, storing the boolean result in `hasEmail`.

Hard

Your Deluge Script

```
hasEmail = student.containsKey("email");
```

Correct! +2.7 points. Great job on Map.

Model answer: `hasEmail = student.containsKey("email");`

Next Question →

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Thamizharasan M

Score: 14.9 /100

Streak: 8

Q 8 / 60

Map

MEDIUM

Retrieve the value for key "name" from the map `student` and store it in `studentName`.

Your Deluge Script

copy / paste / right-click disabled

```
studentName = student.get("name");
```

✓ **Correct!** +1.6 points. Great job on **Map**.

Model answer: `studentName = student.get("name");`

[Next Question →](#)

Answers are checked with flexible keyword/pattern matching against a model solution — this is a learning simulator, not the live Zoho Deluge compiler.

Deluge Script - Basics

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Thamizharasan M

Score: 8.5 /100

Streak: 4

Q 4 / 60

List

HARD

Create an empty List named `rollNumbers` and then add the values 101 and 102 to it.

Your Deluge Script

copy / paste / right-click disabled

```
rollNumbers = List();  
rollNumbers.add(101);  
rollNumbers.add(102);
```

✓ **Correct!** +2.7 points. Great job on **List**.

Model answer: `rollNumbers = list(); rollNumbers.add(101); rollNumbers.add(102);`

[Next Question →](#)

Answers are checked with flexible keyword/pattern matching against a model solution — this is a learning simulator, not the live Zoho Deluge compiler.

Result:

The foundational concepts of the Zoho Deluge scripting language were successfully learned and applied. Through the interactive simulator, all scripting challenges—spanning variables, operators, conditional structures, and data collections—were solved accurately, demonstrating a practical understanding of low-code backend scripting principles.

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